

Factor Books of the Walk-Forward Ablation Ladder

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Abstract

Factor-level analysis of the alpha books produced by four evolutionary research runs on the survivorship-bias-free Nasdaq-100 point-in-time panel (2010–2026): L2 (evolution without retrieval), L4 (knowledge-graph retrieval), L5 (L4 plus an adversarial hypothesis debate) and L7 (L5 plus cross-run memory). Each run evolved for twenty generations under a two-phase progressive reveal; generations 11–20 each first trade one unseen six-month block of 2021–2026 before the population may adapt to it, so the last five years form an honest walk-forward record. All numbers are signal-level information coefficients (IC); no position construction or costs enter.

1 Per-factor performance

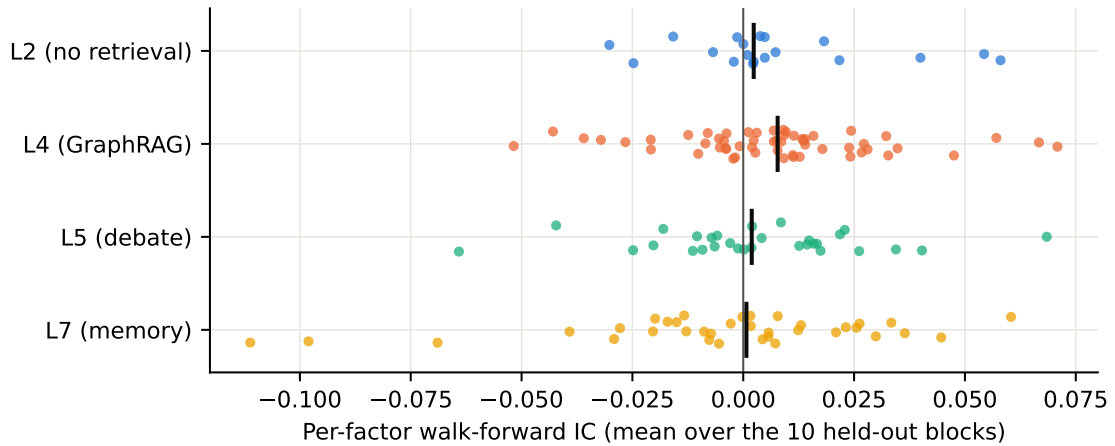


Figure 1: Per-factor IC averaged over the ten held-out walk-forward blocks; the black bar marks the median. L4 produces both the largest book (57 factors) and the highest median walk-forward IC (0.008, 65% positive); the memory arm L7 has the widest left tail.

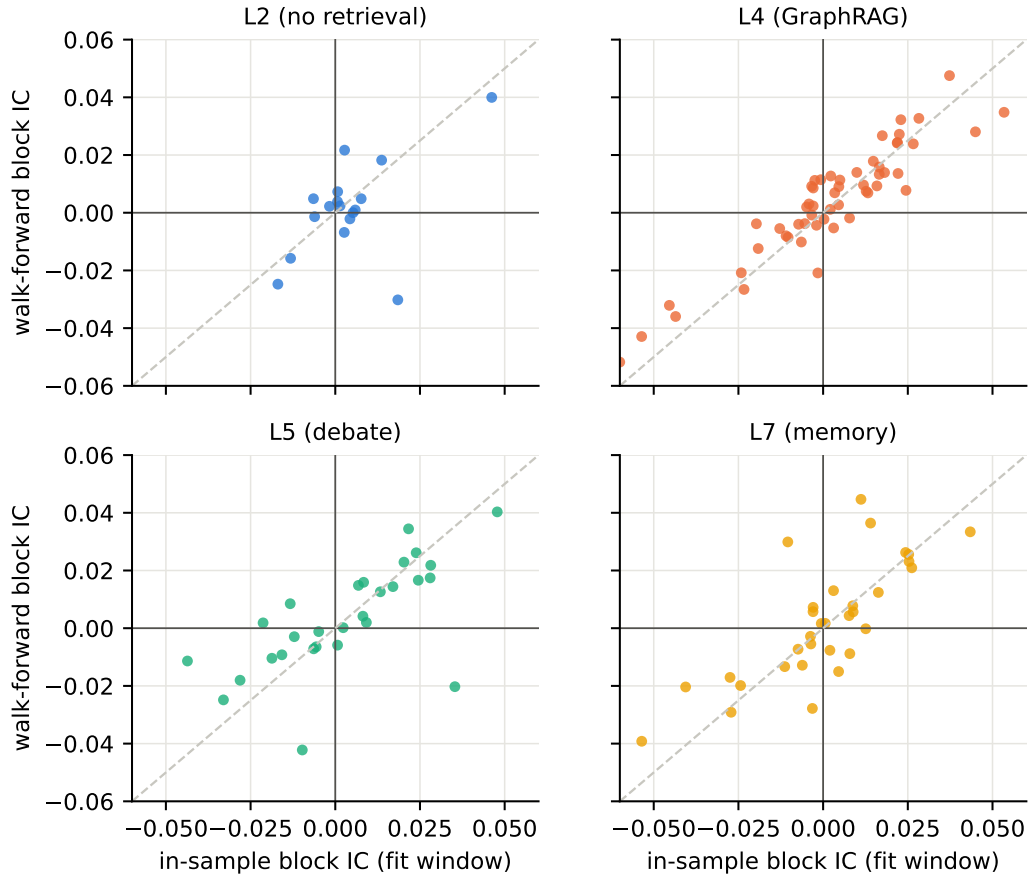


Figure 2: In-sample versus walk-forward per-factor IC (block metric on both sides). Per-factor edge survives out of sample in every arm — mean $|IC|$ is essentially unchanged from fit window to walk-forward — and the rank correlation is high for L2/L4/L7 (0.83–0.94; L5 0.24 due to two outliers).

2 Combining the book into one signal

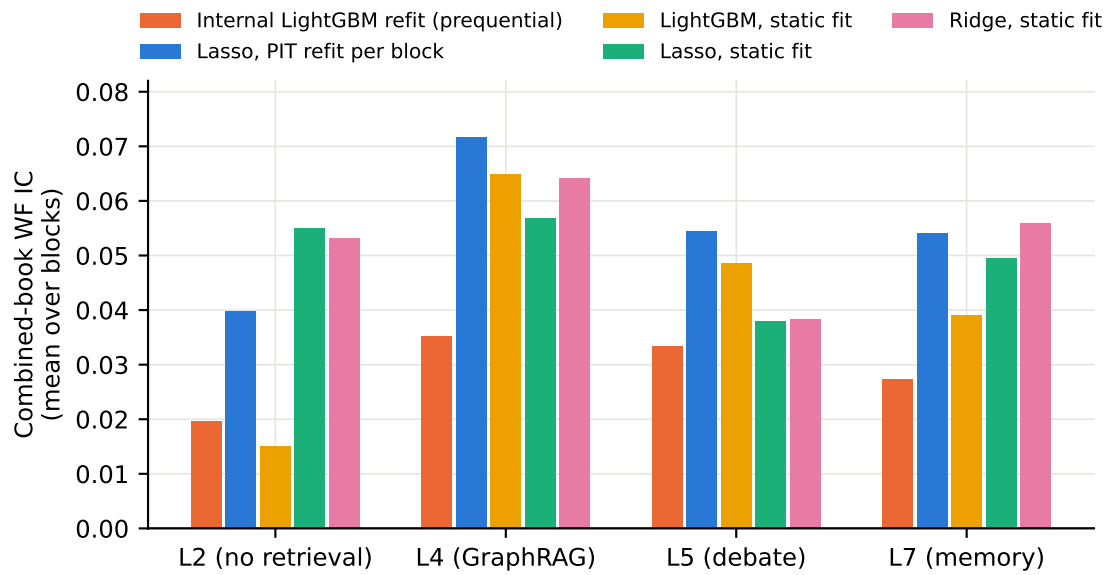


Figure 3: Combined-book walk-forward IC (mean over the ten blocks) for five combiners. The run’s own prequential LightGBM refit — the score the evolution sees internally — is the *weakest* deployment combiner in every arm; a point-in-time lasso refit per block or a simple static ridge adds 50–100%. L4 leads at every combiner (PIT lasso 0.072).

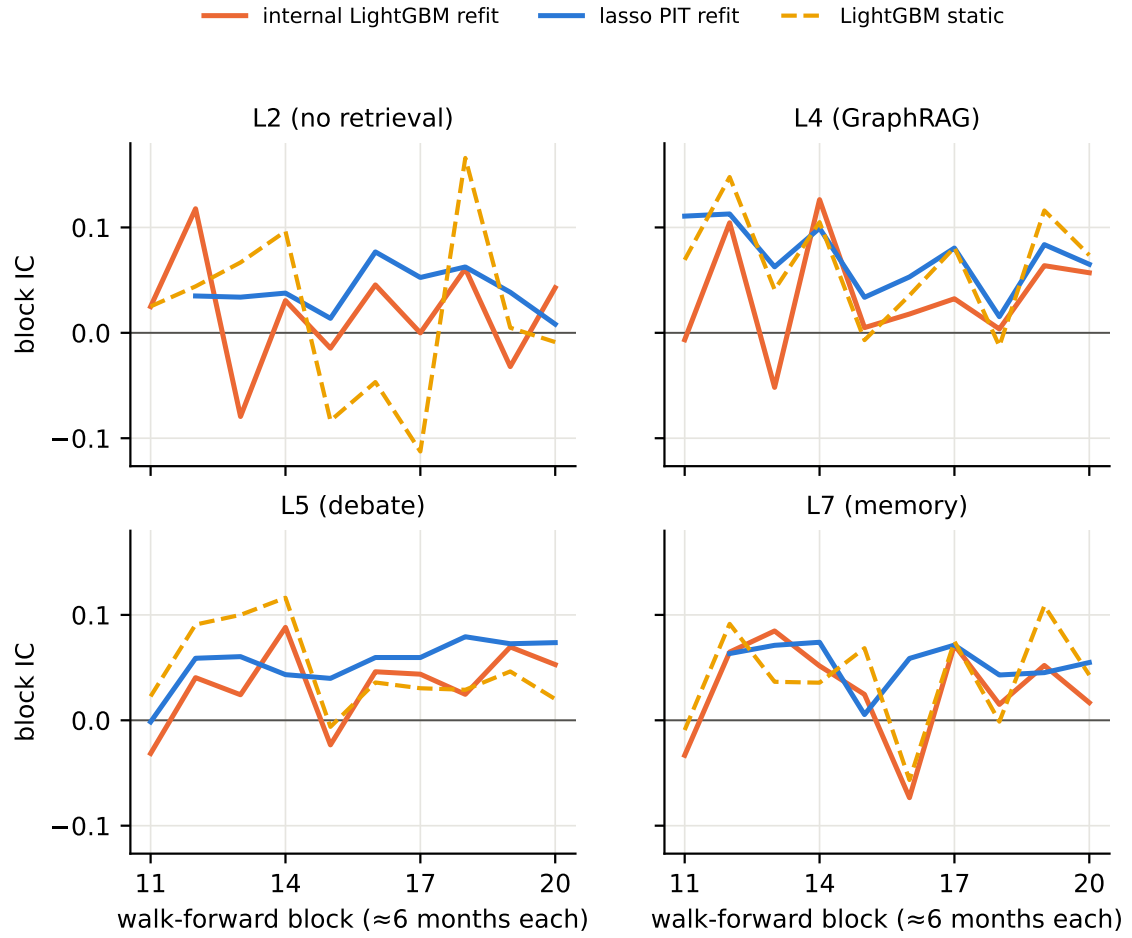


Figure 4: Block-by-block combined IC over the ten held-out blocks (2021–2026). The PIT lasso is positive in 37 of its 38 valid arm-blocks; the tree-based combiners are right on average but far noisier.

3 Archive dynamics over the evolution

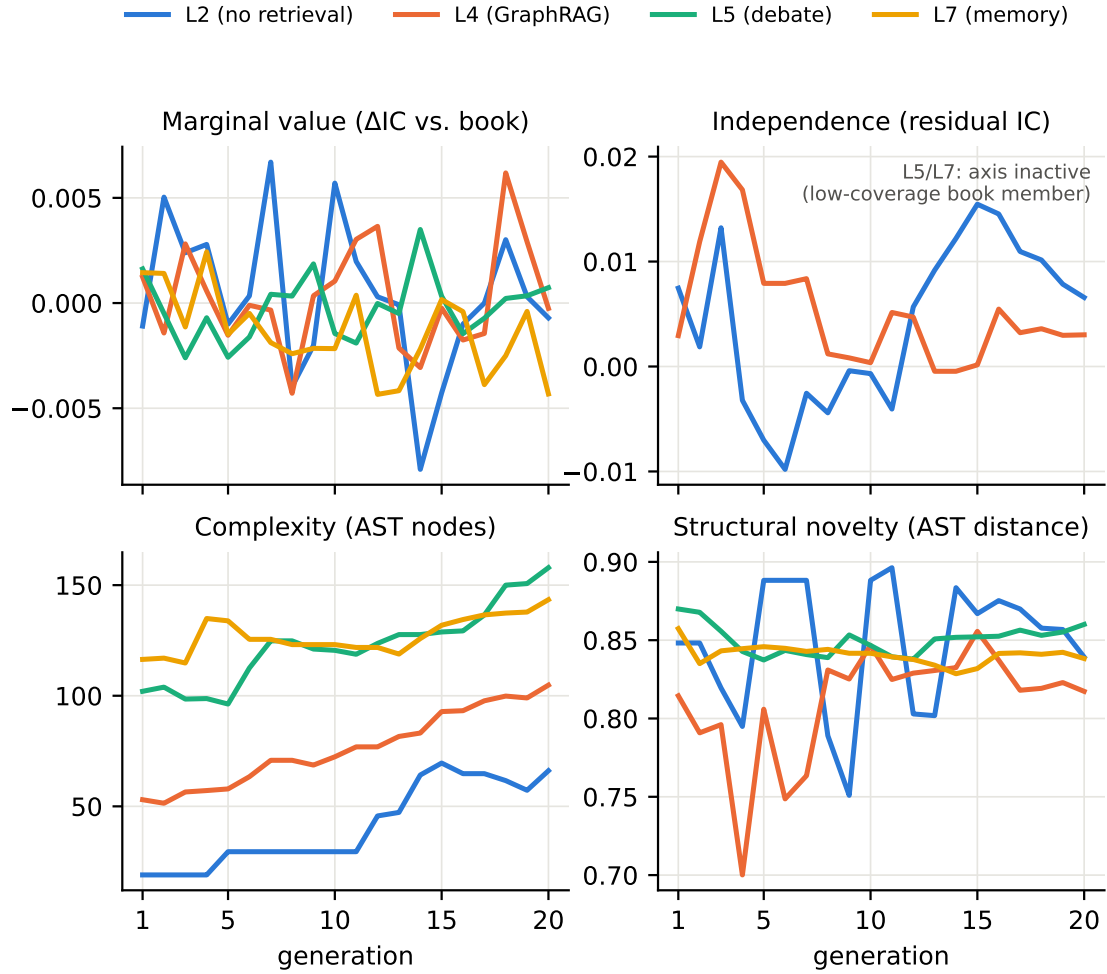


Figure 5: Mean of the four Pareto objectives across the factors that survive into the final archive, per generation (values carried forward between re-scores). Marginal value hovers near zero by construction (each survivor is measured against the rest of the book); complexity ratchets upward in every arm while structural novelty stays flat around 0.85 — the parsimony objective slows but does not stop code growth. The residual-IC independence objective was silently inactive in L5/L7: once a factor with sparse coverage enters the book, the orthogonalisation has too few jointly-complete rows and returns no value.

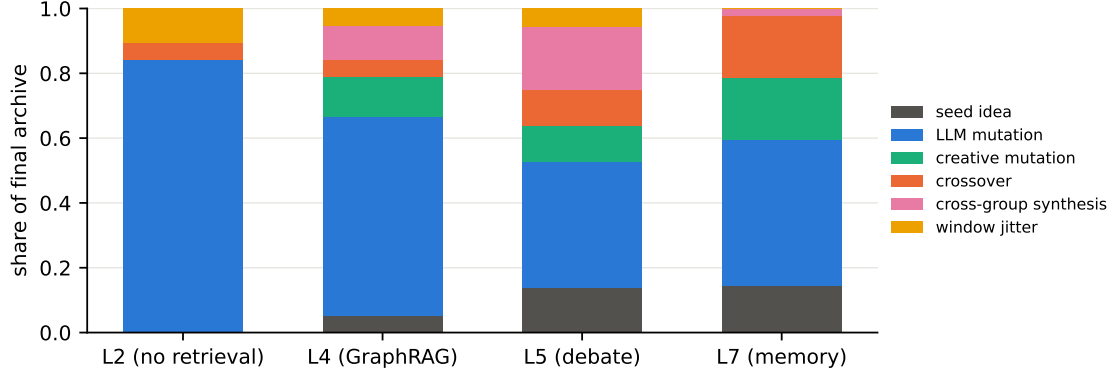


Figure 6: Origin of the surviving factors. Direct LLM mutation dominates everywhere, but the share of survivors from recombination operators (crossover, cross-group synthesis) grows with the richer configurations, reaching a third of the book in L5/L7.

4 Diversity and selection

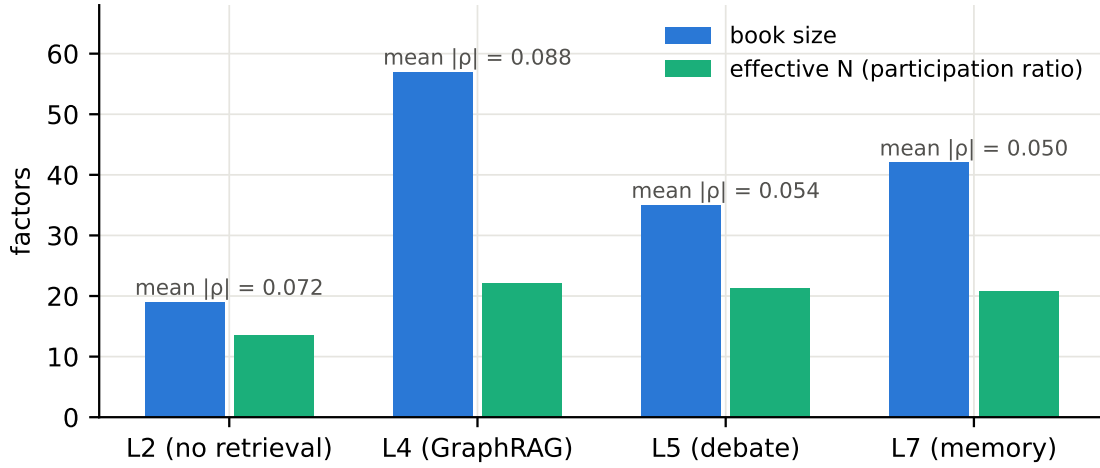


Figure 7: Book diversity via the participation ratio of the signal-correlation eigenvalues, $N_{\text{eff}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$. All evolved books are highly diverse (mean $|\rho| \leq 0.09$); L4’s 57 factors carry about 22 effective independent bets, and the smaller L5/L7 books reach almost the same effective dimension with fewer factors.

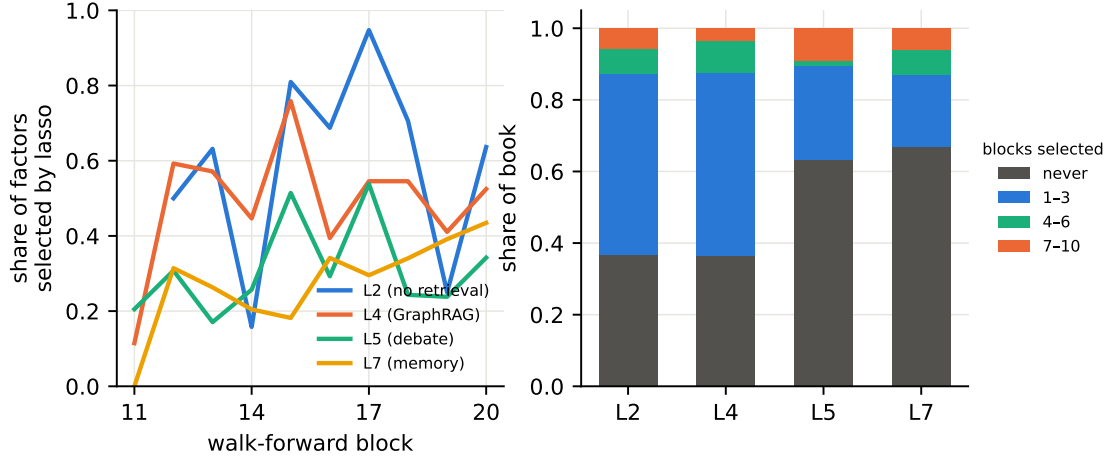


Figure 8: Lasso sparsity under the point-in-time protocol (cross-validated refit before each block). Left: the selected share swings widely between blocks. Right: over the ten refits, 37% (L2/L4) to 63–67% (L5/L7) of factors are never selected, and fewer than 15% are selected in most blocks — yet the combined IC is stable, indicating substantial interchangeability within each book.

Table 1: Cross-validated lasso per phase of the walk-forward period: mean number of selected factors (of available) and mean block IC.

	2021–23	2023–25	2025–26
L2 (no retrieval)	10.5 / 18.5 (.034)	12.3 / 18.8 (.045)	10.0 / 18.3 (.036)
L4 (GraphRAG)	23.7 / 57.0 (.095)	33.3 / 62.8 (.067)	30.3 / 61.0 (.055)
L5 (debate)	9.0 / 39.7 (.039)	15.0 / 37.5 (.051)	10.7 / 39.0 (.075)
L7 (memory)	7.0 / 37.3 (.067)	11.0 / 43.3 (.052)	18.0 / 46.3 (.048)

5 Key observations

1. **Factor-level edge transfers; the combiner is the fragile part.** Mean per-factor $|IC|$ is fully retained from the fit window to the 2021–2026 walk-forward in all four arms, while the combined-model IC depends strongly on the combiner class: simple penalised linear models beat the boosted-tree combiner the search itself uses, in every arm.
2. **Retrieval grounding (L4) is the strongest single addition.** It yields the largest book, the best per-factor medians, the highest effective diversity, and the best combined walk-forward IC at every combiner.
3. **Debate and memory did not pay for themselves at the book level.** L5/L7 books are half positive at the factor level and combine to ~ 0.055 (PIT lasso) versus L4’s 0.072, at comparable diversity.
4. **Selection is unstable, performance is not.** Which factors the lasso picks changes sharply between refits and a third to two-thirds are never picked, yet the combined IC stays positive in 37 of 38 valid arm-blocks — the books are internally redundant in a useful, robust sense.
5. **Two mechanical caveats for the write-up.** Complexity drifts upward over generations despite the parsimony objective; and the residual-IC independence objective silently deactivates once any sparse-coverage factor enters the book (L5/L7 effectively evolved on three objectives), which should be fixed by orthogonalising on pairwise-complete data.